Evolutionary Lability of Sexual Selection and Its Implications for Speciation and Macroevolution

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ABSTRACT: Sexual selection is widely hypothesized to facilitate speciation and phenotypic evolution, but evidence from comparative studies has been mixed. Many previous studies have relied on proxy variables to quantify the intensity of sexual selection, raising the possibility that inconclusive results may reflect, in part, the imperfect measurement of this evolutionary process. Here, we test the relationship between phylogenetic speciation rates and indices of the opportunity for sexual selection drawn from populations of 82 vertebrate taxa. These indices provide a much more direct assessment of sexual selection intensity than proxy traits and allow straightforward comparisons among distantly related clades. We find no correlation between the opportunity for sexual selection and speciation rate, and this result is consistent across many complementary analyses. In addition, widely used proxy variables—sexual dimorphism and dichromatism—are not correlated with the indices employed here. Moreover, we find that the opportunity for sexual selection has low phylogenetic signal and that intraspecific variability in selection indices for many species approaches the range of variation observed across all vertebrates as a whole. Our results potentially reconcile a major paradox in speciation biology at the interface between microevolution and macroevolution: sexual selection can be important for speciation, yet the evolutionary lability of the process over deeper timescales restricts its impact on broad-scale patterns of biodiversity.

Keywords: microevolution, diversification, emergent traits, emergent fitness, comparative demography, sexual dimorphism.

Introduction

The relationship between sexual selection and speciation remains one of the more contentious and poorly understood issues in evolutionary biology. Extensive theoretical and empirical work has explored the ways in which sexual selection might facilitate the speciation process (Lande 1981, 1982; West-Eberhard 1983; Coyne and Orr 2004; Ritchie 2007; Servedio and Boughman 2017; Tinghitella et al. 2018; Calabrese and Pfennig 2020). A corollary from these models is that as the intensity of sexual selection increases in a population, we should observe faster divergence in traits that increase fitness associated with competition for mates.

A number of mechanisms have been proposed to link sexual selection to speciation (Lande 1981, 1982; West-Eberhard 1983; Ritchie 2007; Kraaijeveld et al. 2011; Nosil 2012; Servedio and Boughman 2017; Tinghitella et al. 2018; Calabrese and Pfennig 2020). For example, geographically isolated populations might diverge idiosyncratically in visual or acoustic signals related to mate choice as a result of sexual selection. This differentiation, in turn, might serve as a potent source of premating isolation should these populations come back into contact (West-Eberhard 1983). Regardless of specific mechanisms, phylogenetic comparative methods provide an opportunity to test the effects of sexual selection on speciation rates, and numerous studies have attempted such tests (Barraclough et al. 1995; Panhuis et al. 2001; Coyne and Orr 2004; Rabosky 2016).

Given the difficulty of generating direct and comparable measurements of sexual selection across many species, attempts to investigate sexual selection–speciation connections usually rely on the use of traits as proxies of the intensity of sexual selection (Barraclough et al. 1995; Kraaijeveld et al. 2011; Huang and Rabosky 2014; Servedio and Boughman 2017; Rowe and Rundle 2021). The most widely used traits in this context are mating systems, sexual size dimorphism (SSD; Fairbairn 2007), and clade-specific traits,

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such as coloration in specific feather patches (Beltrán et al. 2021) or dewlaps used as visual displays by lizards (Ng et al. 2017). Previous analyses using such proxy variables have generally found mixed evidence for the relationship between sexual selection and speciation (e.g., bird dichromatism; Kraaijeveld et al. 2011; Cally et al. 2021).

However, proxy variables might have weak or inconsistent relationships with sexual selection, especially when those traits are subject to other constraints or selective factors. For instance, avian plumage dichromatism might be caused not only by sexual selection elaborating male coloration but also by natural selection favoring female crypsis (Wallace 1889; Beltrán et al. 2021). Moreover, dichromatism seems to not correlate either with genomic proxies for sexual selection (e.g., the limited sample in Huang and Rabosky 2014) or with other traits supposedly under sexual selection (but see Price 2015). Other traits commonly used in the comparative literature as proxies for sexual selection also seem to, at best, have a weak association with sexual selection intensity, such as size dimorphism (Fairbairn 2007). Furthermore, in principle trait divergence among populations does not necessarily lead to reduced gene flow or fixation of traits among populations (Calabrese and Pfennig 2020). Thus, for both pragmatic and methodological reasons, macroevolutionary analyses have generally failed to incorporate much of the nuance and complexity embraced by current sexual selection theory.

It has been difficult to apply trait-based proxies across large phylogenetic scales because of dependence on clade-specific traits (such as feather coloration, lizard dewlaps, and antler size/shape, among others). Moreover, the relationship between traits and selection may differ among lineages. For instance, bovid horns and cervid antlers, although similar in some functions, generally differ in development, anatomy, and time availability of the appendage, and the consequences of these differences for macroevolutionary processes remains unknown. Finally, research bias might affect our perception of sexual selection and the modalities by which it operates. For instance, human researchers might be prone to measure visual traits more often than olfactory ones regardless of their relevance for sexual selection, thus underestimating the intensity of selection in an olfactory-oriented species (Ritchie 2007; Kraaijeveld et al. 2011). Consequently, trait-based research on sexual selection and macroevolution may have strong caveats related to the failure to incorporate all (or even the primary) targets of sexual selection (see also supplementary text SI; supplementary texts SI–SVII are available online). Importantly, proxy traits and their potential clade-specific effects may strongly affect conclusions on speciation because most of the phylogenetically independent variation in speciation rates occurs among distantly related clades (Rabosky 2016; see also supplementary text SII). Analyses of traits that are localized to specific clades may thus be limited in power owing to the lack of variation in speciation rate among closely related sets of species (supplementary text SII). As the phylogenetic scale of sampling increases, the amount of phylogenetically independent variation in speciation rate (or any other trait) is expected to increase (fig. S1; figs. S1–S12 are available online). For this reason, some researchers have advocated phylogenetically overdispersed sampling to maximize the power of trait-dependent speciation studies (Rabosky and Huang 2016).

Many of the aforementioned concerns relating to proxy variables would be ameliorated by using direct estimates of skewness in reproductive success within a population (usually known as indices of "opportunity for sexual selection"; see Reed et al. 2023). Indices of opportunity for selection are inherently more comparable across diverse taxa than specific traits because they quantify the demographic signature of sexual selection (e.g., sex-specific variance in mating success) in a taxon-independent manner (see supplementary text SI). In contrast, trait-based analyses assume that the relationship between particular traits and sexual selection (e.g., dichromatism) is generally homogeneous across the focal clade. Here, we collate these indices of sexual selection from population studies of multiple vertebrate taxa and use comparative methods to test whether these indices predict phylogenetic speciation rates. These indices typically consist of variance-based indices for relative mating, fertilization, or reproductive success (examples in fig. 1). Our framework assumes that higher variance in mating success is correlated with the actual intensity of selection, which cannot be guaranteed (Jones 2009; Klug et al. 2010; Marie-Orleach et al. 2021) but is still expected because indices of opportunity for selection represent upper bounds on the linear selection and mating differentials on a phenotype (see more details in supplementary text SI).

We also use this trait-free dataset to directly assess the phylogenetic conservatism of sexual selection intensity (Blomberg and Garland 2002). Surprisingly, we still do not know whether the intensity of sexual selection has phylogenetic signal. On one hand, some traits associated with sexual selection are clearly phylogenetically clustered: for example, antlers in Cervidae, horns in Bovidae (Wilson et al. 2011), sexual signaling in electric fish (Arnegard et al. 2010), and throat and crown color in *Coeligena* hummingbirds (Parra 2010). However, intraclade variation in sexually selected traits is also high, to such a degree that these traits often serve to diagnose or identify distinct species (e.g., in birds), suggesting that such traits are quite labile. Last, we also test the relationship between the intensity of sexual selection and two traits that are commonly used in the macroevolutionary literature as proxies for sexual selection: SSD in vertebrates and dichromatism in birds. Collectively, these analyses provide a first view of the

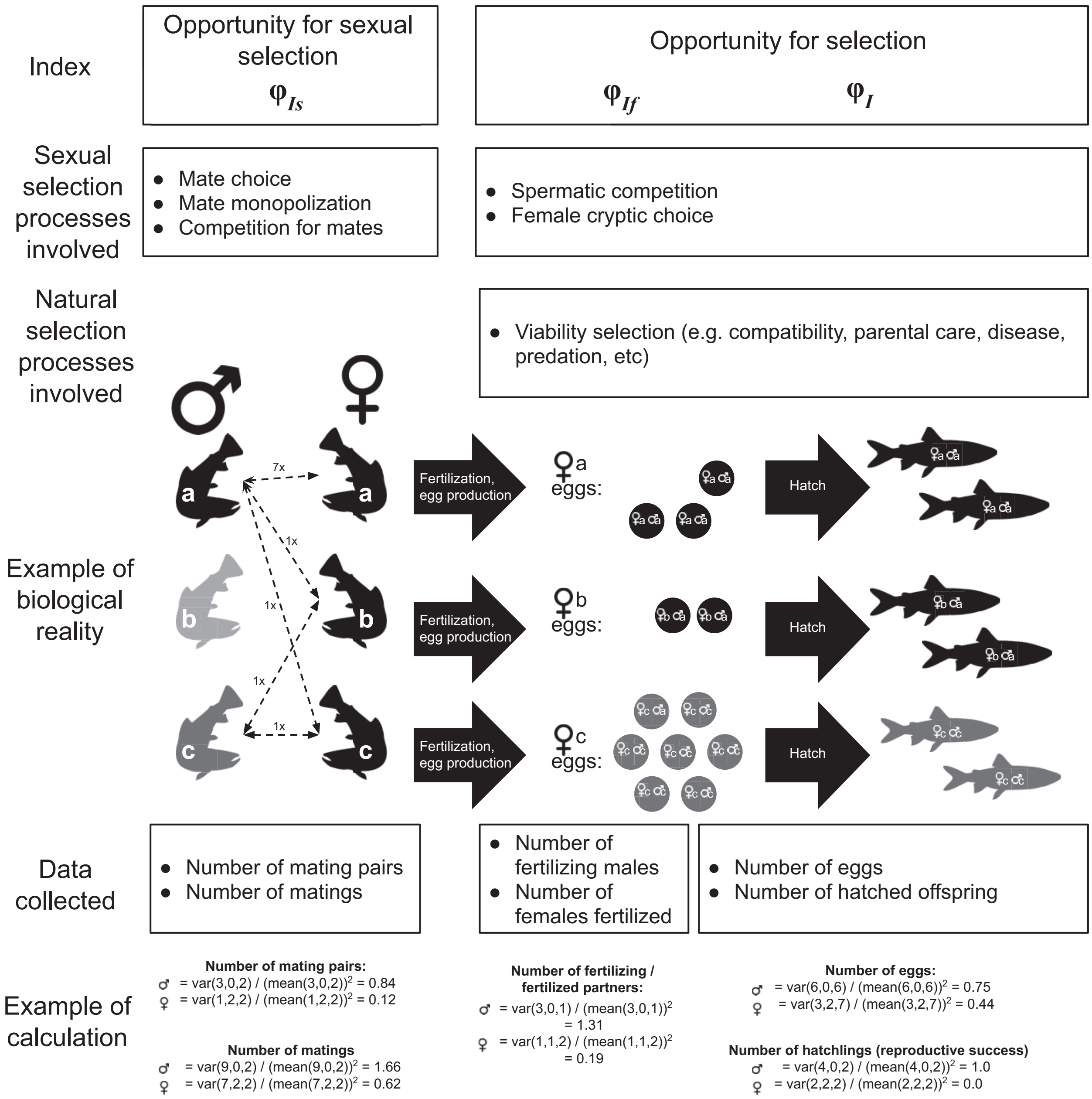

Figure 1: Indices of sexual selection used in this study, including their biological interpretation and relationship with viability selection. In this example, we illustrate how the opportunity for sexual selection (φ_{Is}) can be calculated from the number of pairings (dashed arrows) or the number of copulations (numbers close to each dashed arrow), as well as how the opportunity for selection (φ_{I} and φ_{If}) can be calculated on the basis of the number of eggs or hatchlings. Labels inside eggs and hatchlings show their paternity. Letters a, b, and c show males and females. Note that natural and sexual selection are blurred for φ_{If} and φ_{I} and that the index value varies as a function of data type (pairings or copulation for φ_{Is} and number of eggs/hatched offspring for φ_{I}), sex, and through time. Indices can be sensitive to the inclusion of zero-success individuals (i.e., male b, but note that index values without male b are not shown). Moreover, later selection episodes can change and even revert the net difference in reproductive success (e.g., see females). PhyloPic credits: Chloé Schmidt (CC BY Attribution 3.0 Unported) and Timothy Bartley (CC BY-SA Attribution-ShareAlike 3.0 Unported).

evolution of sexual selection intensity as well as the first empirical, large-scale test of the relationship between selection intensity and widely used proxy traits.

Methods

Opportunity for Sexual Selection Data

Data were obtained through a systematic literature search for studies of nonhuman bony fishes (Osteichthyes) that provided estimates of the intensity of sexual selection (for details, see Macedo-Rego et al. 2024). We later excluded all measurements taken outside field conditions (i.e., laboratory or mesocosm settings). Macedo-Rego et al. (2024) describes the literature search strategy, meta-analytic procedure, data extraction, and filtering (summarized in fig. S2).

All of the opportunity for selection (φ) indices we employed measure the skewness in success (in mating or fertilization, depending on the index used) that may be associated with intrasexual competition for mates (fig. 1; see also supplementary text SI). Opportunity for selection is different from intensity of selection simply because opportunity does not measure change in trait means among generations (Klug et al. 2010; Arnold 2023, p. 12) but appropriately captures differential success among individuals (fig. 1). Opportunity for selection indices also have the additional advantage of not being affected by genetic correlations among traits, which bias estimates of selection intensity (Arnold 2023), while also increasing comparability among distantly related clades (because all clades have populations). We also included Bateman gradients (∇), which measure how correlated are the skewnesses in matings and in offspring number (Bateman 1948). Importantly, all indices are standardized and so can be used for comparing species (Rios Moura and Peixoto 2013). We obtained estimates for the following indices: (i) the opportunity for sexual selection (φ_{Is}), which stands for the variance in matings or mating partners divided by the squared population mean mating success (Crow 1958); (ii) the opportunity for prefertilization sexual selection (φ_{If}), which stands for the variance in fertilization success (e.g., the number of males that fertilized the eggs of a given female or the number of females a given male fertilized) divided by the squared population mean fertilization success (Macedo-Rego et al. 2024); (iii) the opportunity for selection (φ_{I}), which stands for the variance in reproductive success (offspring number) in a given population divided by the squared population mean reproductive success (Wade 1979); and (iv) the Bateman gradient, which stands for the regression slope between individual mating success (i.e., data on φ_{Is}) and reproductive success (i.e., data on φ_{I}; Bateman 1948; Arnold and Duvall 1994).

The main difference between opportunity for sexual selection (φ_{Is}) and the other indices (φ_{If}, φ_{I}, ∇_{ms}, and ∇_{fs}) is that the latter cannot separate the effects of sexual and natural selection (see fig. 1). However, the measurement of sexual selection cannot be simultaneously complete and independent from natural selection, as our only index of opportunity for sexual selection, φ_{Is}, does not provide information either for postmating sexual selection (e.g., sperm competition, cryptic female choice) or for actual reproductive success (see females in fig. 1). Thus, we add φ_{If} and φ_{I} to assess a broader perspective on sexual selection. Note that φ_{If} is relevant because in some species it is extremely difficult to directly determine matings, and authors estimate mating success through genetic paternity analysis. Thus, φ_{If} incorporates the fact that fertilization success does not equate mating success because fertilization also involves postmating sexual selection and natural selection until fertilization takes place (fig. 1) and can therefore bias index values by merging different types of selection (but see Rowe and Rundle 2021). An analogous logic applies to φ_{I}, as it also involves postmating sexual selection and natural selection until reproduction takes place and offspring are assessed by researchers (fig. 1).

We also collected, when available, estimates of the Bateman gradient. The Bateman gradient can be calculated by using estimates of fertilization success in the place of direct mating success (as explained above) as the independent variable. Because Bateman gradients calculated from fertilization success (hereafter, ∇_{fs}; the linear regression of φ_{I} on φ_{If}) or mating success (hereafter, ∇_{ms}; the linear regression of φ_{I} on φ_{Is}) may differ (Marie-Orleach et al. 2016; Cramer et al. 2020), we separated these two types of gradient, acknowledging that (1) Bateman gradients calculated from fertilization success (∇_{fs}) have axes autocorrelated by the influence of the reproductive component of selection in both variables (Ramm et al. 2013) and that (2) the two Bateman gradients capture different information, as ∇_{ms} describes the increase in reproductive success obtained from each unit increase in mating success, while ∇_{fs} shows the fitness return obtained from each reproductive partner. We standardized all Bateman gradient estimates following Jones (2009) and removed negative Bateman gradients from our analysis (details in supplementary text SIII). We $\log_2$ transformed all opportunity for sexual selection indices to improve interpretability (i.e., each unit increment means that the index doubled) across our overdispersed index values. We averaged values of indices by species and sex before transforming. Two species were removed from analyses, as their averaged indices were equal to zero before log transformation: *Anaxyrus americanus* (φ_{Is}; Amphibia: Anura) and *Mandrillus sphinx* (φ_{I}, φ_{If}, and ∇_{ms}; Mammalia: Primates). When the same selection index was collected for the same offspring at multiple developmental stages, we took the earliest measure in an attempt to remove as many natural selection components as possible (see females in fig. 1).

We hereafter refer to individual values of each index that came from some combination of year, article, study, and group-within-study as "populations," not because they refer to demes of relatively unconnected sets of individuals but because those values represent sets of individuals that were able to access each other in the context of each study sampling strategy. Considering all types of sexes and indices we used (i.e., φ_{I}, φ_{Is}, φ_{If}, ∇_{fs}, and ∇_{ms}; see fig. 2), we have a total of 634 selection indices from 288 different populations, as well as a total of 82 species in our dataset (hereafter, "sexual selection dataset").

In all formal analyses (phylogenetic generalized least squares [PGLS], trait model fitting, and measurement of phylogenetic signal; see below), we use the average of populations as species values but perform separate analyses for each combination of sexual selection index and sex. For visualization, we show z-transformed data and "comparative" (i.e., species-level) sample size (hereafter, just "sample size") per index and sex in figure 2. We report in the main text the analyses made with all data available (hereafter, "full dataset"), but we also repeated all analyses after dropping populations in which we were uncertain that zero-success individuals were reported (hereafter, "zero-only dataset"), as the former might be biased to smaller values of sexual selection (see fig. 1). Tree handling, statistical analyses, and plots were made in R version 3.2.2 (R Core Team 2023). Data and analytical scripts are available in the Dryad Digital Repository (https://doi.org/10.5061/dryad.0p2ngf277; Januario et al. 2024).

Traits (Expected to Be) under Sexual Selection

The difference between male and female size (SSD) was operationalized through $\mathrm{SSD} = [(S_{\mathrm{L}}/S_{\mathrm{S}}) - 1]$, where S_{L} is the size of the largest sex, S_{S} is the size of the smaller sex, and the index is arbitrarily multiplied by -1 when males are larger (Lovich and Gibbons 1992; Fairbairn 2007). We used weight (g), snout-vent length (SVL; mm), and body length at maturity (mm) as proxies for size (total of three different SSD operationalizations) and obtained all measurements from the primary literature, only using male and female measurements from the same study to maximize comparability among estimates of sex-specific size. In the full dataset we tested the correlation between weight SSD with four female and five male indices, body length SSD with four female and three male indices, and SVL SSD with four female and four male indices (24 total analyses). In the zero-only dataset we correlated weight SSD with two female and five male indices, body length SSD with one female and three male indices, and SVL SSD with three female and four male indices (18 total analyses). See also supplementary text SV for information on clade-specific analyses.

We obtained bird dichromatism estimates from Armenta et al. (2008), who calculated three different statistics to measure species-specific sexual dichromatism. Two of these assess dichromatism in plumage reflectance (segment classification and color principal component analysis [PCA]), and the other index is the result of a color discriminability model that takes into consideration the signal that is received by the conspecific bird (details in Armenta et al. 2008). In the full dataset we tested for a correlation between (i) color discriminability dichromatism and (ii) four female and four male indices, color PCA dichromatism with three female and four male indices, and segment classification dichromatism with four female and four male indices. In the zero-only dataset, although we could not conduct any analyses involving female indices, we could conduct two analyses for male indices (φ_{If} and φ_{I}).

Tip-Specific Speciation Rates

We used tip-specific speciation rates as an estimate of the present-day instantaneous rate of lineage splitting (Title and Rabosky 2019; Vasconcelos et al. 2022). Speciation rates deeper in time may be difficult to estimate from molecular phylogenies alone, and "recent" (e.g., tip) rates provide the most robust estimate of speciation for macroevolutionary inference (Title and Rabosky 2019; Louca and Pennell 2020; Vasconcelos et al. 2022), provided assumptions about the completeness of taxonomic sampling are met (Chang et al. 2020; Freeman and Pennell 2021). Tip rates were estimated by Cooney and Thomas (2021) using BAMM (hereafter, λ_{BAMM}; Rabosky et al. 2013; Rabosky 2014) and the DR statistic (hereafter, λ_{DR}; Jetz et al. 2012). Tip rates were estimated from recently published time-calibrated vertebrate phylogenies (i.e., Jetz et al. 2012; Tonini et al. 2016; Faurby et al. 2018; Jetz and Pyron 2018; Rabosky et al. 2018) and methods that account for clade-specific taxonomic undersampling. We $\log_2$ transformed tip rates for downstream analyses.

Three species present in the sexual selection dataset were missing from the speciation rate dataset—*Chiroxiphia lanceolata* (Aves: Passeriformes), *Empidonax trailli* (Aves: Passeriformes), and *Xiphophorus birchmanni* (Osteichthyes: Cyprinodontiformes); in these cases, we used the average of the tip-specific speciation rate for the corresponding genus of the missing species instead. This is justified by the fact that speciation rate estimates are generally autocorrelated among closely related sets of species, as they likely share the same rate shifts along most of their evolutionary history and because the inference procedures use information from multiple adjacent branches within a phylogeny to derive an overall estimate (see fig. 7

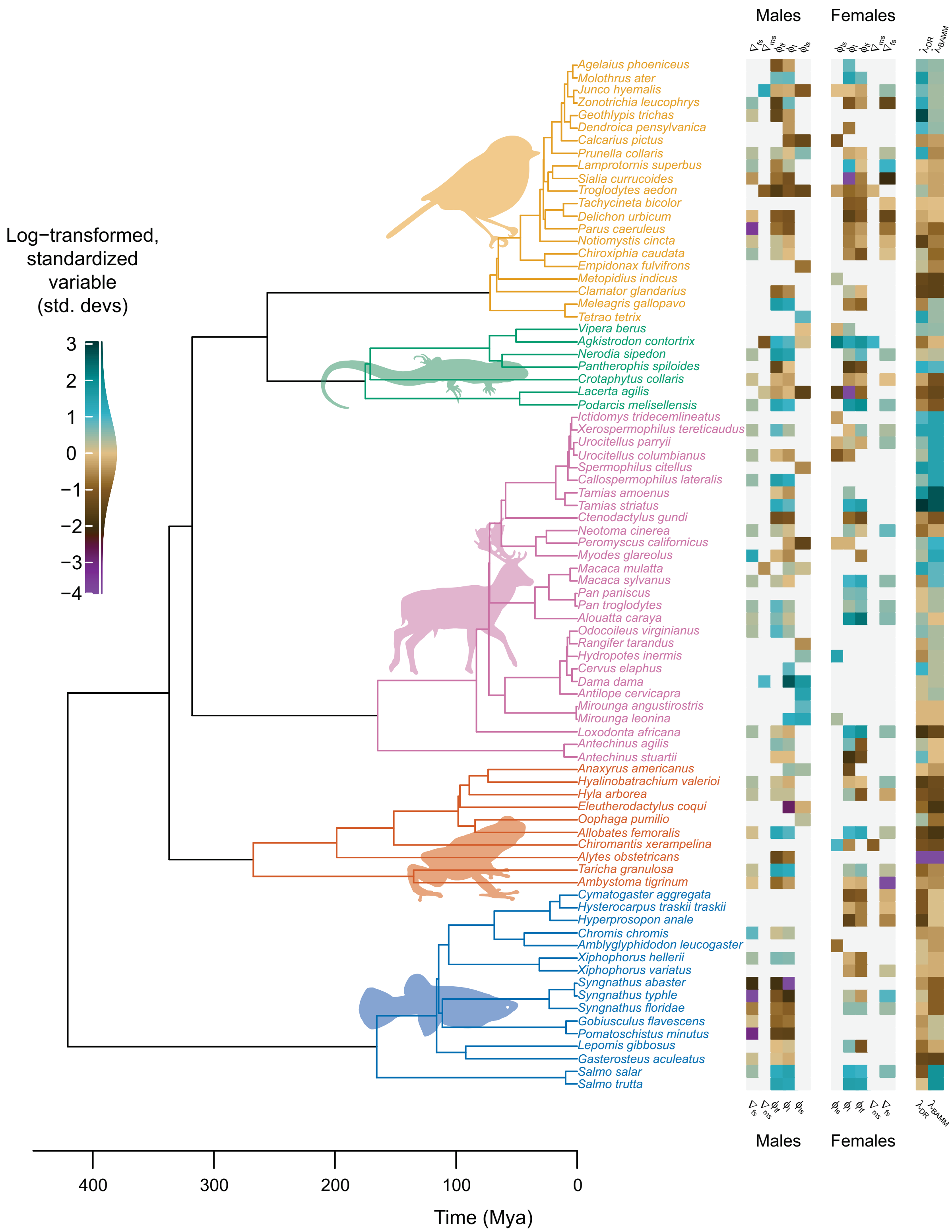

Figure 2: Standardized, species-averaged indices of sexual selection (opportunity) in a phylogenetic framework. Tile colors show sexual selection indices and speciation rates across vertebrate clades and sexes (full dataset). In this scale, similar values (i.e., colors) should be aligned if there is a positive correlation with at least intermediary levels of R^2 among variables—for example, as among the same type of index (i.e., φ and ∇) for the same sex (figs. S3, S4) or as λ_{BAMM} and λ_{DR} (where $R^2 = 0.4627$). φ_{Is} = opportunity for sexual selection; φ_{If} = opportunity for prefertilization sexual selection; φ_I = opportunity for selection; ∇_{ms} = mating-based Bateman gradients; ∇_{fs} = fertilization-based Bateman gradients. Colored branches are from time-calibrated phylogenies of the five major vertebrate clades illustrated. All variables were z transformed. Silhouette credits are provided in the acknowledgements. Mya = million years ago.

in Title and Rabosky 2019). In total, we were able to compare four female indices and five male indices with λ_{DR} or λ_{BAMM} in each of the full and zero-only datasets (total = 36 analyses; see also supplementary text SV for information on clade-specific analyses).

Phylogeny

We collated maximum clade credibility trees of ray-finned fishes (Rabosky et al. 2018; Chang et al. 2019), amphibians (Jetz and Pyron 2018), squamates (Tonini et al. 2016), birds (backbone from Cooney et al. 2017; clades from Jetz et al. 2012), and mammals (Upham et al. 2019), dropping all species without DNA data. When no maximum clade credibility tree was provided in the references above, we constructed it using phytools (Revell 2024) and 100 random trees from the posterior or bootstrapped sets of trees provided. We also created a backbone for the five groups listed above (black branches in fig. 2) using crown ages from Benton et al. (2015), to which we grafted each subclade phylogeny (using ape [Popescu et al. 2012], phangorn [Schliep 2011], and paleotree [Bapst 2012]). For species present in the sexual selection dataset but missing from the DNA-based tree, we randomly chose one species within the same genus as a replacement to keep most of the relevant topology of the tree (fig. 2). We used this reduced tree in each PGLS and phylogenetic signal analyses described below, always dropping tips without data for each particular analysis. Tables S1–S5 (tables S1–S9 are available online) show sample size (i.e., number of species) per analysis, and figure 2 illustrates the data distribution along the vertebrate tree.

We do not address phylogenetic uncertainty in our analyses because the molecular trees we are using (i.e., those with species from the sexual selection dataset only) differ only slightly—for instance, the maximum Robinson-Folds distance among 100 sampled subtrees of the amphibian tree is equal to 2 (results not shown). Presumably, this reflects the fact that most of the phylogenetic uncertainty is associated with shallow nodes. Moreover, the original source trees have—at best—limited accounting for true phylogenetic uncertainty (e.g., because they are conditioned on a single underlying dataset, often highly enriched for mitochondrial DNA).

PGLS Analysis

We used PGLS to test the relationship between indices of sexual selection and tip speciation rates. We also used PGLS to assess the suitability of sexual selection proxies commonly used in the literature. The proxy traits used are (1) SSD in vertebrates and (2) sexual dichromatism in birds. As is the case with other empirical ways of measuring sexual selection in a comparative framework (see above), it is possible that our selection indices do not capture relevant aspects of sexual selection. Thus, to evaluate the consistency among our selection indices we used PGLS to cross-correlate all indices of sexual selection.

We evaluated PGLS using the whole vertebrate dataset (hereafter, "vertebrate-level analyses"), but the relationship among selection indices and speciation rate and/or proxy traits may be clade specific. To account for this possibility, we did additional analyses where clade (amphibian, bird, squamate, mammal, ray-finned fish) is a covariable modulating the effect of the opportunity for sexual selection on speciation or proxy traits. We repeated our PGLS analyses, including clade as a factor, with automatic stepwise model selection as implemented in phylolm (Tung Ho and Ané 2014) to select the combination of variables (index and/or clade, or null) best supported by the data. As an additional step to explore clade-specific patterns, we redid these PGLS analyses including only species within the same clade (hereafter, "clade-level analyses"). Importantly, these clade-level analyses are expected to have substantially lower statistical power than the analyses that use all of the vertebrate lineages in our dataset because they are based on far fewer species. In spite of their reduced power, we feel that these analyses can indicate trends and patterns that are clade specific and that may be obscured by the analyses at larger scales and may also help identify variation in effects across clades that potentially compromise interpretations of the full dataset. For both taxonomic levels (vertebrate and clade specific) we ran all PGLS combinations among selection indices and speciation rates or proxy traits, but some did not have invertible matrices and so were excluded (tables S1–S4). We fitted models using caper (Orme et al. 2023; except the automatic stepwise model selection) and simultaneously optimized the phylogenetic signal and the PGLS parameters (Revell 2010). We report parameter P values and model ΔAICc (Akaike information criterion corrected for small sample size) in tables S1–S4.

We did not formally correct for multiple comparisons in our analyses; rather, we assessed significant results in light of overall model fit, especially with regard to whether results were consistent among different types of operationalizations. For example, we considered further only those results that were consistent across both measures of speciation rate or that showed consistent patterns by sex and/or across all five major vertebrate clades. We also conduct power analyses (see below) to help guide our interpretations.

Power Analyses

We performed a simulation analysis to assess the power of our dataset and analysis framework to recover a true correlation among indices of sexual selection and traits (in our case, speciation rates, dimorphism, and dichromatism). We used the R package mvtnorm (Mi et al. 2009) as well as each PGLS-specific phylogeny to simulate 1,000 traits under a Brownian motion (BM) model (and Pagel's Λ equaling 1) that has a known correlation with the response variable (tip speciation rates or traits). We drew correlations from a uniform distribution between -1.0 and $+1.0$ and registered the proportion of simulated datasets in which PGLS recovered a statistically significant correlation (i.e., the slope's P value is smaller than .05) for that pair of phylogeny and tip rates. A similar framework was applied before in a study of speciation rate (Singhal et al. 2022).

In our first power analysis, we evaluated the capacity of a given species set to recover a true nonnull result (i.e., avoid a type II error). We modeled this as a binary simulation result (i.e., PGLS is significant or not) following a binomial model with $k = 1$ and probability of significance

$$P_{\text{signif}} = \frac{1 + \exp(-r(1 - t))}{1 + \exp(-r(\rho - t))},$$

where r is only an adjustment parameter (a "rate of increase"). The "threshold" parameter t represents the value of true correlation that, if present, has at least a 50% chance to be detected, given a phylogeny and the tip values of the dependent variable (speciation rate or trait). If t is larger than 1, this indicates that the relative odds of a type II error (not detecting a true correlation) are always large for that set of species and phylogeny and so that particular analysis has no power (="powerless"). Finally, ρ is the module of the simulated correlation (see graphical explanation in fig. S9). We estimated parameters through maximum likelihood (were likelihood $= \prod P_{\text{signif}}$, obtaining parameter values for r and t for each pair of phylogeny and tip rates evaluated in all of our empirical analyses.

In our second power analysis, we estimated the range of predictor values (selection indices and speciation rates) that could have generated a R^2 similar to what was observed. To do this, we departed from an original set of simulations conducted in our first power analysis referring to a single empirical PGLS. We then subsetted these simulations, keeping only those with an estimated R^2 that deviated less than 0.1 from the empirical correlation value for that analysis. This process was repeated for all sets of empirical analyses.

Finally, in supplementary text SIV we compare the power of our analyses with the power that would be achieved in family-level analyses of vertebrates. Our aim is to explicitly compare the statistical power in our phylogenetically overdispersed set of taxa with the power of typical clade-specific comparative analyses of sexual selection that would include most species within a taxonomic family (e.g., Bro-Jørgensen 2007; Ng et al. 2017; Beltrán et al. 2021). Taxonomic families are sometimes chosen by researchers because species within the same family share the same proxy traits and/or because they are a frequent target for dedicated phylogenetic studies. Thus, family-level analyses may more appropriately encode a supposed effect of sexually selected traits on speciation, regardless of the statistical power that exists in said analyses to recover true relationships with speciation rate. In our evaluation of vertebrate families, we included all 577 vertebrate families that have less than 250 species (species counts based on the species included in the speciation rate dataset, not on "true" family richness). We estimated family-specific statistical power to detect true relationships with speciation using the same methods described above (see additional details in supplementary text SIV).

Phylogenetic Signal of Sexual Selection

To quantify the phylogenetic signal of sexual selection in each index, we evaluated Blomberg's K (Blomberg et al. 2003) and Pagel's Λ (Pagel 1999) using phytools (Revell 2024). Table S5 also shows phylogenetic signal measured with different metrics (Moran 1950; Abouheif 1999), which are model-free measurements of phylogenetic signal whose significance was assessed with permutation among tips (Münkemüller et al. 2012). We also compared the empirical values of Blomberg's K of all selection indices with values for a set of reference traits across vertebrates, as originally compiled by Blomberg et al. (2003). These reference traits include aspects of morphology, ecology, and physiology and provide context for interpreting the observed phylogenetic signal in selection indices. As a complement (see "Discussion"), we also compared the AICc of the following models of trait evolution: BM, early burst (EB), single-peak Ornstein-Uhlenbeck (OU), and white noise (WN)—a model in which traits show no phylogenetic signal. We fit those four models to all indices, sexes, and traits using the package geiger (Harmon et al. 2008; Pennel et al. 2014), with any model with ΔAICc smaller than 2 (relative to the smallest AICc) considered equally good in explaining our data. We plotted results using base R packages (R Core Team 2023): ggplot2 (Wickham et al. 2016), ape (Popescu et al. 2012), and phytools (Revell 2024).

To confirm our phylogenetic signal findings were not an artifact of the atypical taxon sampling in our tree (e.g., clusters of closely related species separated by large phylogenetic distances), we performed a positive control, testing whether we would estimate significant phylogenetic signal for traits that are a priori expected to show such signal within our species pool. We tested the phylogenetic signal of body size as measured by the average mass of both sexes, $\log_2$ transformed across the set of species in the sexual selection dataset. We then compared this estimate to the level of phylogenetic signal previously reported in the literature for this trait in many vertebrate clades. To the extent that our estimates of phylogenetic signal in sexual selection are unbiased by our sampled topology, we expected to observe (1) a significant phylogenetic signal in body size and (2) a Blomberg's K value that is close to the median of Blomberg's data for body mass across many vertebrate clades. We also measured "local" estimates of phylogenetic signal to test whether phylogenetic signal differed among subclades in the vertebrate tree. The details, results, and discussion of this analysis are shown in supplementary text SVI.

Results

Consistency among Indices

Log-transformed averages of φ_{Is}, φ_{If}, and φ_{I} are pairwise positively correlated for each sex (figs. S3, S4), except for male φ_{Is} and φ_{If}, male φ_{Is} and ∇_{ms}, and male φ_{If} and ∇_{ms} (fig. S3). We also found that Bateman gradient values are biased toward smaller slope values when the R^2 of their linear regression is low (fig. S5). Intraspecific variation in opportunity indices is very large, and single species with multiple measurements overlap greatly with the range of values observed in all vertebrates (fig. 3).

Sexual Selection and Speciation Rate

In the vertebrate-level analysis, there is generally no correlation between indices of opportunity for sexual selection and phylogenetic speciation rates (tables S1, S2). For the full dataset, this lack of correlation was observed in all 10 analyses made with male indices and in all eight analyses made with female indices. In the analyses made with the zero-only dataset, no significant correlations were observed in any of the eight analyses with female indices, and nine (out of 10) analyses were nonsignificant in the male dataset (but ∇_{fs} significantly correlates with λ_{BAMM}). Figure 4 shows the relationship among λ_{DR} and selection

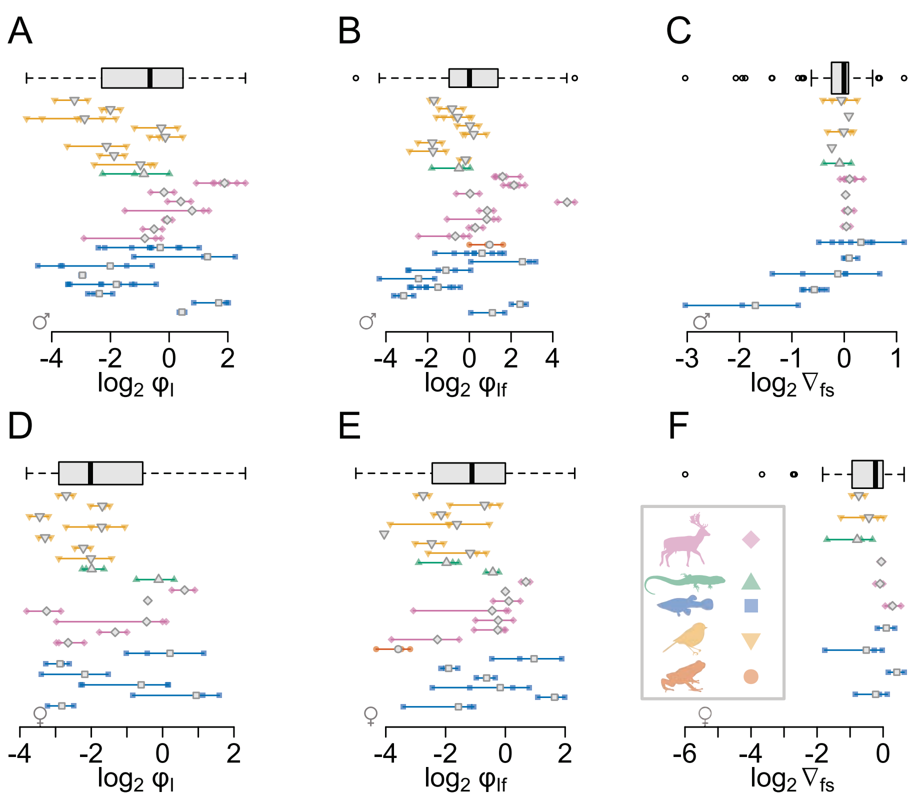

Figure 3: Intraspecific variation in opportunity for sexual selection across indices. Horizontal bars represent within-species range, and smaller points represent population-level observations. Gray central, outlined points represent species means. Gray boxplots at the top of each panel show the entire distribution across all vertebrate populations for the corresponding panel. The range of variability within individual species is so large that many species span much of the entire range of values seen across all vertebrates. Point shapes and colors refer to vertebrate clades. Single-valued species are present only in the boxplots. Lines do not match among panels. Figure S8 shows a complete version of this plot that includes all indices.

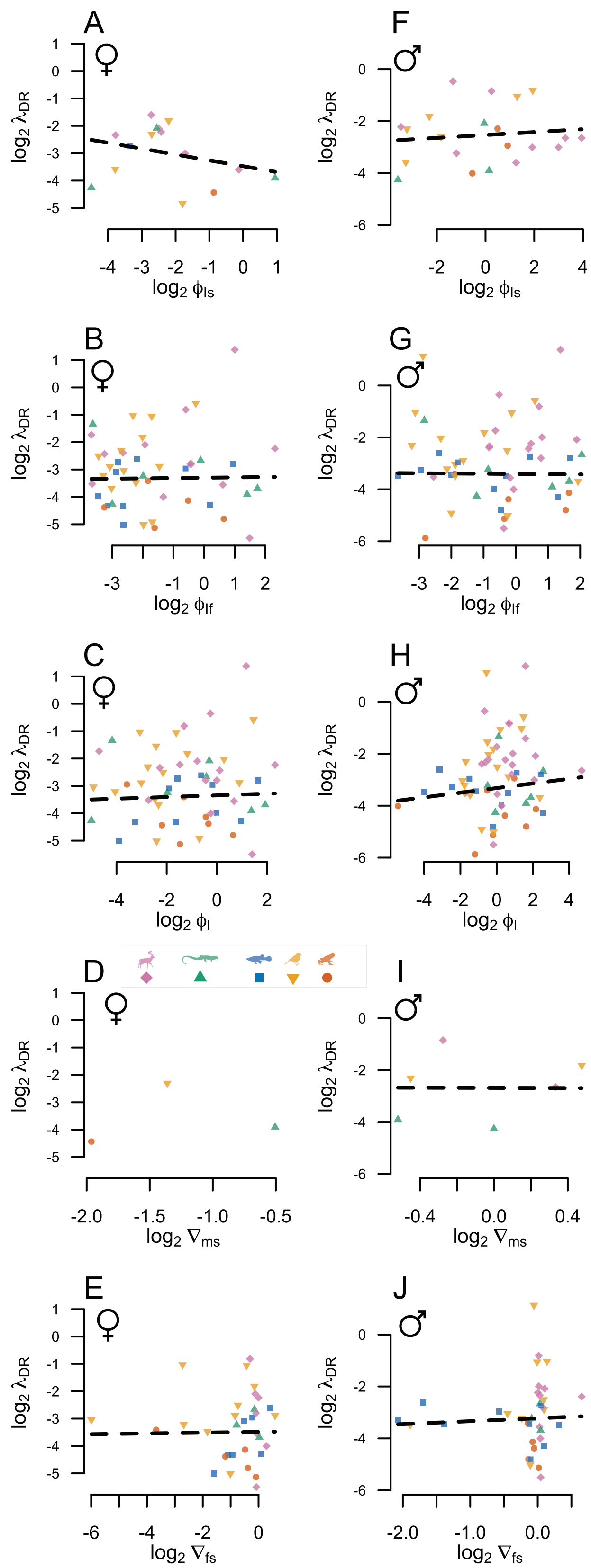

Figure 4: No relationship between indices of sexual selection (opportunity) and speciation rates (λ_{DR}). Points show species-sex-specific averages. None of the results illustrated (all from the full dataset) have slopes that significantly deviate from zero (nonsignificant phylogenetic generalized least squares relationships shown with black dashed lines). We could not fit a model with λ_{DR} as a function of female ∇_{ms} because of the number of points (n = 3; *D*, no line shown). Point shapes and colors refer to vertebrate clades. Red circle = amphibians; yellow downward triangle = birds; blue square = fishes; pink diamond = mammals; green upward triangle = squamates. Parameter estimates are shown in table S1.

indices in the full dataset (separated by sex; see also fig. S6 for results referring to λ_{BAMM}). Clade-specific analyses are shown in supplementary text SV.

In summary, no significant relationship found between speciation rate and selection index is consistent between measurements of speciation rate, across clades, and across sexes. This qualitative result remains unchanged if models are evaluated with ΔAICc instead of slope *P* value (tables S1, S2, S6). The automatic stepwise model selection mostly selected the null model in both datasets, and when it selected a nonnull model, the number of parameters significantly different from zero was never larger than two (see the material on Dryad [https://doi.org/10.5061/dryad.0p2ngf277; Januario et al. 2024]).

Proxy Traits and Opportunity for Selection

In the vertebrate-level analyses SSD correlates consistently (among datasets) only with female φ_I (negative correlation with mass SSD) and with male φ_I (negative correlation with SVL SSD). The three other proxies that correlate significantly in the full dataset (out of 24 PGLs evaluated) are not consistent with the zero-only dataset (tables S3, S6; see also fig. S14). Similarly, the other two significant PGLS (out of 18) made with the zero-only dataset were not consistent with the full dataset (tables S4, S6).

In summary, consistent results, when they happen, occur as negative correlations between φ_I and SSD, and consistency occurs between zero-only and full datasets (table S6). Moreover, the only vertebrate clade that follows the vertebrate-level pattern is amphibians (and only in the full dataset of males). All other (few) significant results we found present no consistency between proxy trait, sex, selection index, or taxonomic level. Results are overall the same if models are evaluated with ΔAICc instead of slope *P* value (tables S3, S4, S6). Stepwise model selection generally selected the null model for both datasets, and when it selected a nonnull model, the number of parameters significantly different from zero was never larger than two (see material on Dryad [https://doi.org/10.5061/dryad.0p2ngf277; Januario et al. 2024]).

Phylogenetic Signal in Sexual Selection Opportunity

Most combinations of indices and sexes have low phylogenetic background (= whole tree) signal (fig. 5; table S5), a pattern that becomes more evident when we compare the estimated values with other vertebrate traits previously reported in the literature or even with the phylogenetic signal of the proxy traits evaluated here (fig. 5). The main exception to this pattern of low phylogenetic signal is male φ_I (reproductive success variance divided by the squared mean reproductive success) and female φ_{If} (same as φ_I but success is measured by the number of reproductive partners instead of offspring number)—both of which present significant Blomberg's K (table S5). Proxy traits present nonsignificant phylogenetic signal in all its operationalizations except SSD in weight, which has significant Pagel's Λ (table S5). Results vary only slightly when phylogenetic signal is measured by Moran's *I* or Abouheif's *C* (table S5). In our tests for heterogeneity in phylogenetic signal among clades (local indicator of phylogenetic association [LIPA] analyses) passeriform birds seem to more frequently deviate from this background phylogenetic signal, but this result is not consistent across indices and sexes (see details in supplementary text SVI; fig. S12).

When we performed our positive control—testing whether our phylogenetic sampling would reveal phylogenetic signal in traits expected to show it—we found that body mass has significant phylogenetic signal in all metrics (table S5), and corresponding Blomberg's K values are very close to other estimates of phylogenetic signal in traits not associated with sexual selection (fig. 5). Thus, we see no evidence that our phylogenetic sampling (fig. 2) affects our ability to detect phylogenetic signal when it is expected to be present.

In terms of trait evolution modeling, the WN and OU models were generally selected—or were equally as good as other models—in all evaluated traits (table S5). Notable exceptions from this pattern are female ∇_{ms}, which is equally well modeled by OU and EB models; PCA-based dichromatism, which is equally well modeled by BM and WN; and body size, which is equally well modeled by OU and BM.

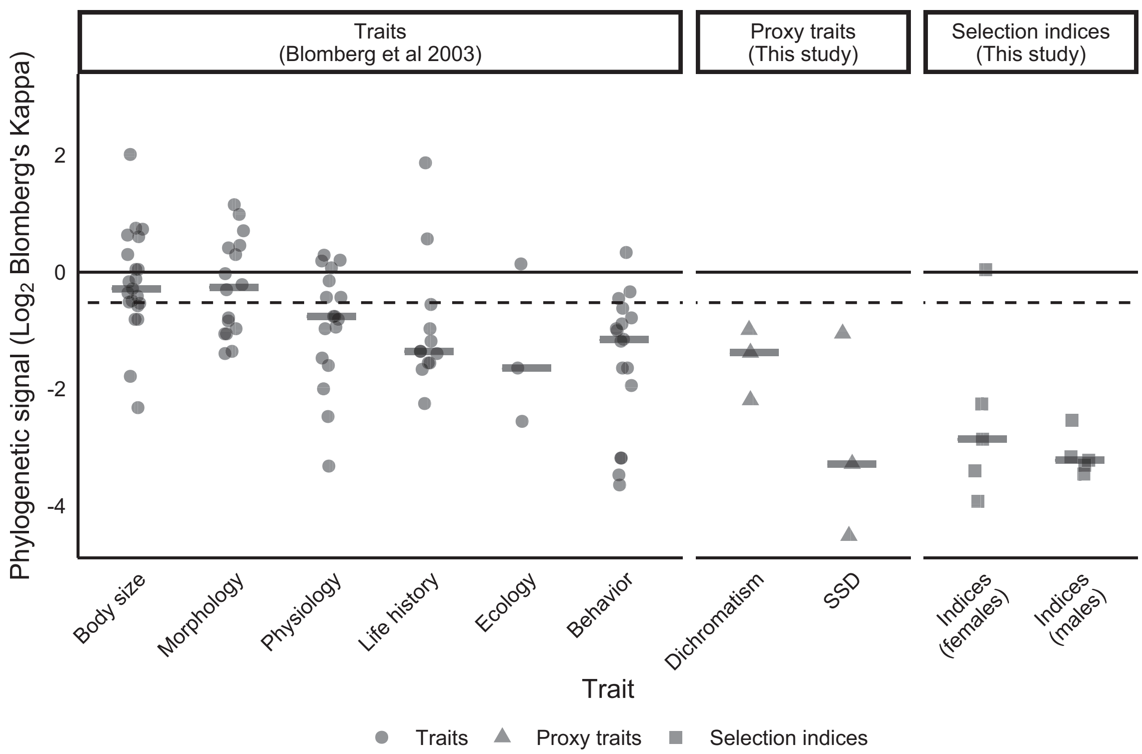

Figure 5: Phylogenetic signal in the opportunity for selection (squared points), for traditional sexual selection proxies (triangles), and for a reference set of traits from Blomberg et al. (2003) that span a range of organismal functions (circles). Horizontal thicker gray lines show corresponding median. The continuous line that crosses all panels shows the phylogenetic signal expected under Brownian motion, and the dashed line shows phylogenetic signal in body weight (average of the two sexes) calculated using the same data compiled for the dimorphism analyses and the phylogeny used in all analyses of this study. K estimates are log transformed for visualization purposes. Estimates from Blomberg et al. (2003) span the same sets of clades considered in this study; a breakdown of points by major taxon is shown in figure S10.

Power of PGLS Analyses

When we analyze the distribution of simulated correlations that could have generated our empirically observed correlation (second power analysis), all quantiles compromising 95% of the simulated R^2 include zero. This means these datasets do not provide enough evidence to reject either (1) a value of true zero correlation or (2) that the true correlation might actually have the opposite signal than the observed. However, our first power analysis shows that most PGLS with more than 20 species had R^2 thresholds smaller than 0.3 (fig. S7), for analyses evaluating both speciation (tables S1, S2) and proxy traits (tables S3, S4). This means that true correlations whose R^2 is medium or large are likely to be detected in our dataset. Although power is smaller in the zero-only dataset and in the clade-level analyses (tables S1–S4; figs. S7, S13), these results suggest that our approach has, broadly speaking, reasonable power to detect true correlations in most of the better-sampled vertebrate analyses—provided the correlation is at least moderately strong.

Importantly, the power of analyses that span all five clades of vertebrates (in both full and zero-only datasets) was typically greater than the power that a "typical" family-specific comparative dataset would generate (figs. 6, S13; supplementary text SIV). The increased power of the phylogenetically overdispersed analyses results from the inclusion of much more phylogenetically independent variation in speciation rates at large phylogenetic scales than is typically available at the family level or below (fig. S13; see also supplementary text SIV). Clade-specific analyses had considerably smaller power (figs. S7, S13).

Discussion

The extent to which population-level characteristics are associated with macroevolutionary speciation rates remains a critical and unresolved problem in biodiversity science. Here, we demonstrate that population-level measurements of the opportunity for sexual selection are characterized by high evolutionary lability and intraspecific variation across vertebrate lineages. This result implies that—at least for the clades considered here—sexual selection has sufficient variability and at sufficiently fine-grained phylogenetic scales such that it is unlikely to have an appreciable impact on speciation rates over macroevolutionary timescales. Consistent with this result, we also found a demonstrable lack of correlation between speciation rate and the opportunity for sexual selection. The strength of our test and the generality of our results arise from several factors: (1) the concordance of results across several hierarchical levels (PGLS, phylogenetic signal, intraspecific variation); (2) the broad scope of our analyses,

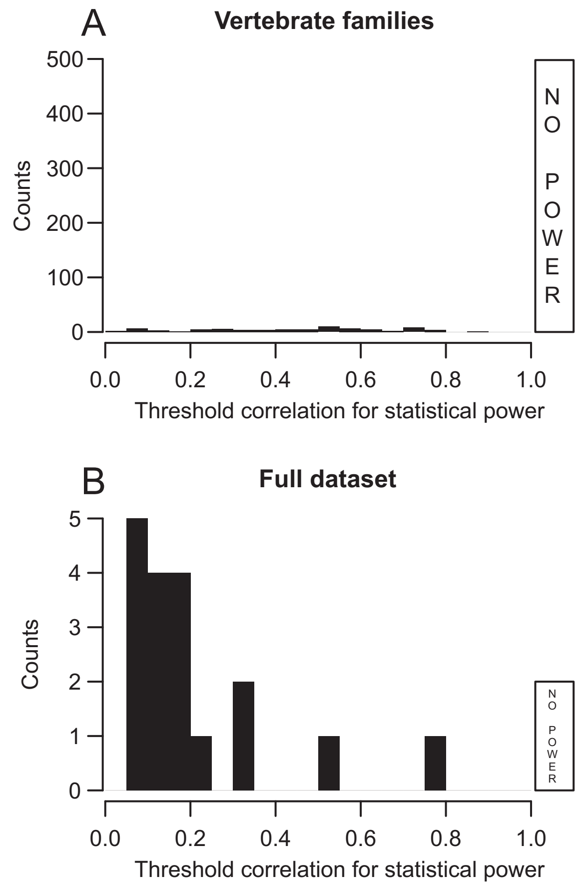

Figure 6: Statistical power in "typical" (*A*; see supplementary text SIV) and phylogenetically overdispersed (*B*) analyses of speciation rate. The black histograms show the number of analyses in different power bins (i.e., threshold parameter in our first power analysis), while the white bars show the number of analyses with no power (out of the scale in the *x*-axis but still on scale in the *y*-axis). Only 13.7% of the family-level analyses have at least some power, and it is generally small (ordered quartiles of the analyses with power: 25% = 0.257, median = 0.461, 75% = 0.600). This contrasts sharply with the analyses in the major dataset used in our analyses ("full dataset"), where 90% of analyses have some power, and it is skewed toward smaller correlation thresholds (ordered quartiles: 25% = 0.090, median = 0.153, 75% = 0.220). Figure S13 shows a complete version of this plot that includes all clade-level analyses and the zero-only dataset.

which encompass numerous evolutionarily shifts in speciation rate; and (3) the comparability of our focal traits across vertebrates, specifically the indices of the intensity of (opportunity for) sexual selection.

Phylogenetic Scale and Macroevolutionary Analysis

Although it may look surprising that we can confidently reject a strong association between sexual selection and speciation using relatively small phylogenetic datasets

($N < 100$), we show that the power of our tests is generally much greater than the statistical power found in family-level analyses that include twice as many species in fully sampled clades (figs. 6, S13; supplementary text SIV). This is important because clades with a few hundred species represent the sample size used for many family-level, trait-centered comparative analyses of sexual selection (Bro-Jørgensen 2007; Ng et al. 2017; Beltrán et al. 2021) and of speciation rate (Kraaijeveld et al. 2011; Cally et al. 2021; Anderson and Weir 2022; Helmstetter et al. 2023). Perhaps counterintuitively, our results support previous calls for fundamental shifts in the nature of taxon sampling for comparative diversification analyses (Rabosky and Huang 2016). Most studies of trait-dependent diversification emphasize comprehensive clade-wide sampling that effectively restricts the amount of phylogenetically independent variation in both the trait and the diversification rates. Here, we had greater statistical power using sparse but phylogenetically overdispersed taxon sampling than we generally would have had with dense taxon sampling of any individual family-level vertebrate clade (fig. 6). The low power found in most of the 507 family-level clades explored (fig. 6) suggests an acute need to expand the scope of comparative diversification analyses by using quantitative metrics of evolutionary and ecological processes that can be compared across greater phylogenetic scales than is typically possible with clade-specific proxy variables.

However, as the phylogenetic scale of analysis increases, it is also true that other differences between clades may influence any general relationship between a specific trait and speciation rate. For example, sexual selection might consistently influence speciation rates, but only relative to clade-specific baseline rates, which might differ systematically among groups as different as frogs, flounders, and finches. Thus, there is a fundamental tension between the need to increase scale (more independent rate shifts) and the corresponding increase in uncontrolled sources of variation as phylogenetic scale increases (e.g., baseline rates, clade-specific factors). We have no clear resolution for this general phenomenon in macroevolutionary studies other than to suggest that a priori determination of a "just-right" scale could be helpful for maximizing the power of future comparative tests (fig. 7).

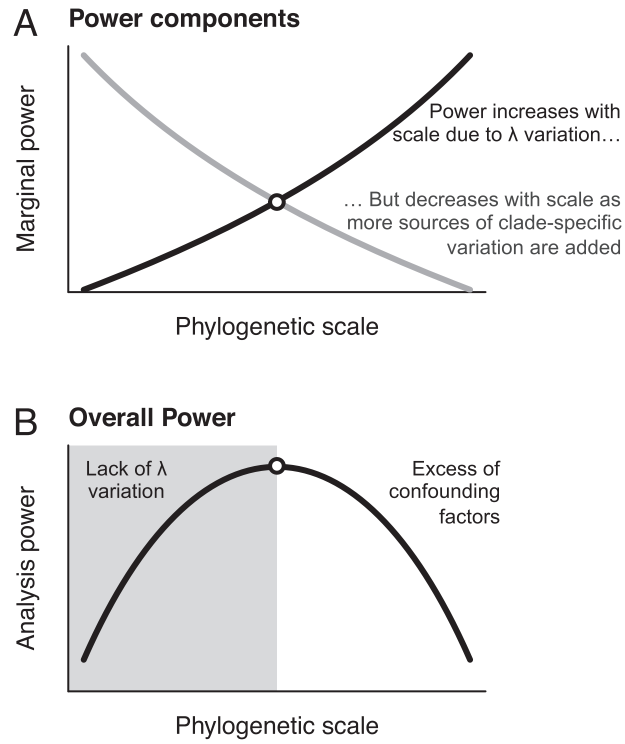

Figure 7: A trade-off between factors that increase and decrease statistical power with phylogenetic scale leads to a "Goldilocks zone" for phylogenetic comparative analyses. *A*, Overall statistical power should increase with phylogenetic scale (black line) because larger phylogenetic scales (e.g., large clades) encompass more phylogenetically independent variation in diversification rates. However, power can also decline with increasing scale because larger clades contain more biological heterogeneity and thus more uncontrolled sources of variation (gray line). *B*, The combined effect of these two trends is to maximize statistical power at some intermediate scale. The confounding factors in *B* could come from many sources, including shifts in clade-specific "background" rates of speciation as well as from clade-specific heterogeneity in the relationship between the focal traits and speciation.

Intraspecific Variability in Sexual Selection

Many of the species in our dataset with multiple population-level observations have intraspecific ranges of selection indices that span a substantial fraction of the full range observed across all vertebrates (fig. 3). This result is somewhat aligned with the literature, which has reported variation in those indices through time and among populations (van Daalen and Caswell 2019; Marie-Orleach et al. 2021). This large amount of variation is not limited to a few species or clades, based on the data we have at hand (figs. 3, S12). It is unclear whether the opportunity for selection is a unique type of trait or whether other types of demographic metrics are equally variable (figs. 4, 5). Important developments on this front would include measuring the magnitude of stochastic variation in these indices under natural conditions (i.e., how much of this variation comes purely out of chance events that occur in specific years or breeding seasons) as well as establishing whether intense sexual selection leaves consistent patterns in genetic data (e.g., as in Huang and Rabosky 2014). These are all different questions related to the same endeavor: to determine how and why the intensity of sexual selection (or at least the opportunity for sexual selection) varies so much within species.

Sexual selection is expected to vary in space due to density-dependent effects or due to the distribution of resources associated with mating (Machado et al. 2016; Muniz and Machado 2018; Tsuji and Fukami 2020). Furthermore, there is growing evidence of spatial variation in the intensity of sexual selection—for example, *Syngnathus* pipefish (Rispoli and Wilson 2008; Mobley and Jones 2009), *Xiphophorus* fish (Tatarenkov et al. 2008), *Geothlypis* warblers (Taff et al. 2013), wrens (*Troglodytes*; LaBarbera et al. 2012), and *Clamator* cuckoos (Bolopo et al. 2017). Interestingly, spatial variation in SSD and some other proxy traits was shown in some vertebrate lineages (DeGregorio et al. 2018; Tarr et al. 2019; Chelini et al. 2021), although the generality of this pattern remains unknown. Determining how much of this variation corresponds to latitudinal gradients, climatic clines, genetic differences among populations, or qualitative changes in habitat (Boughman 2001; Seehausen 2006; Wagner et al. 2012) is an open empirical question and if confirmed must later be reconciled with our findings showing intermediate phylogenetic signal in proxy traits.

The Phylogenetic Signal of Sexual Selection

Across most indices we considered, sexual selection was characterized by minimal phylogenetic structure, such that closely related species were no more likely to have similar values than distantly related species. This low phylogenetic signal provides a first glimpse of the phylogenetic structure of the opportunity for sexual selection across vertebrates. Importantly, the estimates of phylogenetic signal we obtained for our sexual selection indices are much lower than those observed for other types of traits (e.g., the traits from Blomberg et al. 2003) and are even different from the traits we evaluated here: SSD, dichromatism, and average body size (fig. 5). Given this evidence, it is not surprising that OU and WN models fit most data better (table S5) than EB or BM, as together these models can capture a spectrum of processes that have little or no phylogenetic signal. Although inferring microevolutionary process from phylogenetic signal alone is problematic (Hansen and Martins 1996; Revell et al. 2008), we note that intra- and interspecific patterns found are fully consistent across our dataset, as we observe both high intraspecific variability and low conservatism across the phylogeny.

Sexual Selection and Proxies

Our results suggest no consistent correlation between opportunity for sexual selection and two commonly used proxy traits for the intensity of sexual selection: SSD and color dichromatism in birds (13 significant results out of all 137 proxy trait analyses employed in all datasets; tables S3, S4, S6; fig. S14). Many comparative analyses use proxy traits to estimate aspects of sexual selection, and thus our collective understanding is tied to the limitations of this practice. However, in contrast to our results on the opportunity for sexual selection, the proxy traits we examined (SSD and dichromatism) show intermediate phylogenetic signal compared with other traits and selection indices (fig. 5). These results suggest that the same lability that decouples sexual selection's intensity from speciation rates may also influence the connections between selection intensity and traits used as proxies for sexual selection "strength."

One could point to an apparent paradox: how can proxy traits for sexual selection have phylogenetic signal if direct measurements of sexual selection (as measured by opportunity indices) imply that it is quite labile? We see no necessary conflict. For instance, consider a runaway-like process, where genetic covariances create positive feedback loops between female preference and male trait (Lande 1981, 1982). In this scenario, trait divergence will stop only when genetic variation is depleted or when natural selection counterbalances sexual selection. As a consequence, it is not the intensity of sexual selection that drives how divergent traits can be but the evolutionary pressures that counterbalance it—a reasoning already discussed partially (Lande 1981) but that is still generally neglected in the macroevolutionary literature on sexual selection. Scenarios like these, where different factors limit divergence due to sexual selection, can potentially explain mismatches between selection intensity and trait values and are congruent with our results on phylogenetic signal. This hypothesis would also make it less surprising to observe that males of monochromatic species like the white-crowned sparrow have a more skewed reproductive success (larger opportunity for selection) than males of highly dichromatic species like the red-winged blackbird—and that females present opposite patterns (see fig. S14). The discrepancies in correlation/phylogenetic signal between proxy traits and selection indices found in our study raise the question that maybe the replenishment of additive genetic variation, the structure of the genotype-phenotype map, or even the alignment between sexual and natural selection may be the determinants of long-term divergence, even in traits under strong sexual selection.

Although SSD and dichromatism may not be condition dependent (Peters et al. 2011), other sexually selected traits are (e.g., armament size). This raises questions about the informativeness of such traits in comparative studies, especially when used as proxies for evolutionary processes. With that said, condition (i.e., environmental) dependence itself (as well as other aspects of the genotype-phenotype map) might be an evolutionarily labile trait (Curlis et al. 2017; Rowe and Rundle 2021) and could ultimately determine the extent of divergence in response to many factors other than the intensity of sexual selection (e.g., life history or ecological context; Taff et al. 2013; Morehouse 2014; Tentelier et al. 2016; Rowe and Rundle 2021). From this perspective, it appears to us that using traits as proxies for evolutionary processes is a fragile construction. Indeed, we are not aware of any article claiming to use trait values as proxies for the intensity of natural selection. Why, then, is the literature so accepting of claims that specific trait values can denote the intensity of sexual selection?

Sexual Selection and Speciation

We did observe a single significant correlation between selection opportunity and speciation rates (out of 35), and some nonnull models were preferred by the automatic stepwise model selection procedure (supplementary text SV). However, we interpret these results with skepticism, as they were generally inconsistent among simple subdivisions of our dataset (tables S1–S4, S6). Importantly, even if the intensity of sexual selection does not contribute to species richness differences among major clades of vertebrates, our results are agnostic to the importance of sexual selection to speciation and/or to the evolution of reproductive isolation. Current speciation theory conceptualizes it as a noninstantaneous process that requires not only the evolution of reproductive isolation between demes but also their genetic and ecological differentiation, as well as the persistence of these demes throughout geological time (Allmon 1992; Seehausen 2006; Dynesius and Jansson 2014; Allmon and Sampson 2016; Harvey et al. 2019). Consequently, any of these different components of what is required for speciation could set the rate-limiting step of the speciation rate as measured in molecular phylogenies (Rabosky 2016), meaning that sexual selection can be extremely important for the evolution of reproductive isolation but still does not predict which clades will actually speciate faster because it is only one part of the speciation process. For instance, sexual selection could be causally creating shallow-scale divergence among demes, but if these demes do not persist through geological timescales, they will not contribute to the differential clade-level richness heterogeneity. Moreover, our results align with recent assessments suggesting that sexual selection, even when strong, does not necessarily lead to speciation (Servedio and Boughman 2017; Mendelson and Sanfran 2021; Shuker and Kvarnemo 2021; Rowe and Rundle 2021; Boughman and Servedio 2022). Nonetheless, it should also be noted that other population-level proxies that were hypothesized to correlate with speciation in the noninstantaneous view discussed above, such as the evolution of reproductive isolation (Rabosky and Matute 2013; Freeman et al. 2022) or of genetic structure (Singhal et al. 2018, 2022; Nitschke et al. 2018; Burbrink et al. 2023; Wacker and Winger 2024), also have no predictive power for speciation rates. Still, other large-scale investigations (Ellis and Oakley 2016; Janicke et al. 2018) have found an association between traits/selection indices and clade richness in animals as a whole, raising the possibility that sexual selection affects large-scale diversity patterns through its effects on extinction rates, and we discuss this specific connection in supplementary text SVII.

Limitations

The selection indices used here (φ_{I}, φ_{Is}, φ_{If}, ∇_{fs}, and ∇_{ms}) are not free from limitations. For example, they depend on the number of individuals competing (Klug et al. 2010; Jennions et al. 2012), they can vary stochastically across years or breeding seasons (van Daalen and Caswell 2019; Marie-Orleach et al. 2021), and other indices capture more tightly the variation in selection intensity that is due to mating systems (Henshaw et al. 2016). Moreover, variance in mating is irrelevant to selection unless it is aligned with variance in reproductive success (Jones 2009; Henshaw et al. 2016; see females in fig. 1) and reproductive success also evolves by natural selection (Holsinger 2013). This is important because φ_{I}, φ_{If}, and ∇_{fs} are particularly underestimated if offspring is inviable or predated before researchers have access to it (Bergeron et al. 2012; Ramm et al. 2013; Cramer et al. 2020). Finally, field context or study design might also affect measurement accuracy, especially in the case of φ_{Is} (Klug et al. 2010; Cramer et al. 2020; but see Henshaw et al. 2016). This long caveat list reflects the fact that sexual selection is a multifaceted evolutionary process that is generally difficult to measure, not necessarily from the specific methodology used here (see fig. 1). With that said, owing to our meta-analytical procedure to collect data, our indices represent the current understanding of vertebrate variation in sexual selection opportunity. Moreover, most of our opportunities for selection (φ) and our Bateman gradients (∇) are largely consistent among themselves when cross correlated within the same sex (figs. S2, S3). This consistency suggests that our data capture a relative rank of the intensity of sexual selection among vertebrate species.

Some of the analyses employed have low sample sizes, but our overall conclusions remain robust because they are based not on particular tests but on the collective behavior of the analyses, with emphasis in the more powerful analyses that involved all vertebrate species (supplementary text SIV). As a reviewer commented, comparative analyses commonly report relatively low R^2 values; as such, finding that our analyses would generally have had power to reject the null hypothesis only when the true correlation exceeded 0.3 may suggest triviality to our main results. However, as we show in our analyses, some commonly used traits are—at best—noisy proxies for sexual selection. These same proxies have been used in numerous studies of sexual selection and speciation (Kraaijeveld et al. 2011; Cally et al. 2021; Helmstetter et al. 2023), often at greatly restricted phylogenetic scales where power would likely be even lower on account of the reduced variability in speciation rate associated with such clades (figs. 6, S1, S7, S13; supplementary text SIV; Maddison and FitzJohn 2015; Rabosky and Goldberg 2015; Rabosky and Huang 2016). Thus, the ambiguous results or low R^2 reported in the literature (Kraaijeveld et al. 2011; Cally et al. 2021; Janicke and Fromonteil 2021; Winkler et al. 2024) might result from the imperfect/noisy nature of widely used proxy variables or from the limited taxonomic scope of many previous investigations (e.g., noisy proxy or low phylogenetically independent variability in speciation rate).

Our analyses have additional limitations that are common to most studies of phylogenetic speciation rates, including any systematic biases across clades in divergence time estimation (Bromham and Cardillo 2019) and effects of taxonomic over-/undersplitting (Chang et al. 2020; Freeman and Pennell 2021). Importantly, our analysis cannot reject the possibility that sexual selection is correlated with speciation within taxonomic families of vertebrates that were not included in our dataset; such families necessarily include many taxa in which sexual selection has been studied extensively but where comparable opportunity or selection indices were not available. It is also possible that most traits have marginal impacts on speciation rates for reasons relating to underlying biology itself, including clade-specific shifts in other variables that also affect evolutionary rates (Anderson et al. 2023; Helmstetter et al. 2023). Thus, finding which clades, if any, have remarkably different relationships between sexual selection and speciation represents a much more involved and nuanced question that remains empirically open.

Conclusions

The relationship between sexual selection and broader biodiversity patterns remains unclear. Here, we showed that (i) the opportunity for sexual selection is uncorrelated with speciation rate or with traits commonly used as proxies for sexual selection, (ii) the opportunity for sexual selection has extremely low phylogenetic signal, and (iii) there are large amounts of intraspecific variation in the opportunity for sexual selection. Our findings indicate that we still have much to learn about which factors (e.g., environmental, stochastic, or demographic components) control the variance in sexual selection intensity, especially how this variation impacts both trait divergence and speciation rates. Our results, although surprising, can help explain why so many phylogenetic diversification studies have reported ambiguous or conflicting signals with respect to the relationship between macroevolution and sexually selected traits.

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Statement of Authorship

Project conceptualization: M.J. and D.L.R. Experimental design: M.J. and D.L.R. Data analysis, coding, simulations: M.J. First draft: M.J. and D.L.R.. Data collection/data interpretation: R.C.M.-R. All authors contributed to revisions and data visualization.

Data and Code Availability

Data and code have been archived in the Dryad Digital Repository (https://doi.org/10.5061/dryad.0p2ngf277; Januario et al. 2024).

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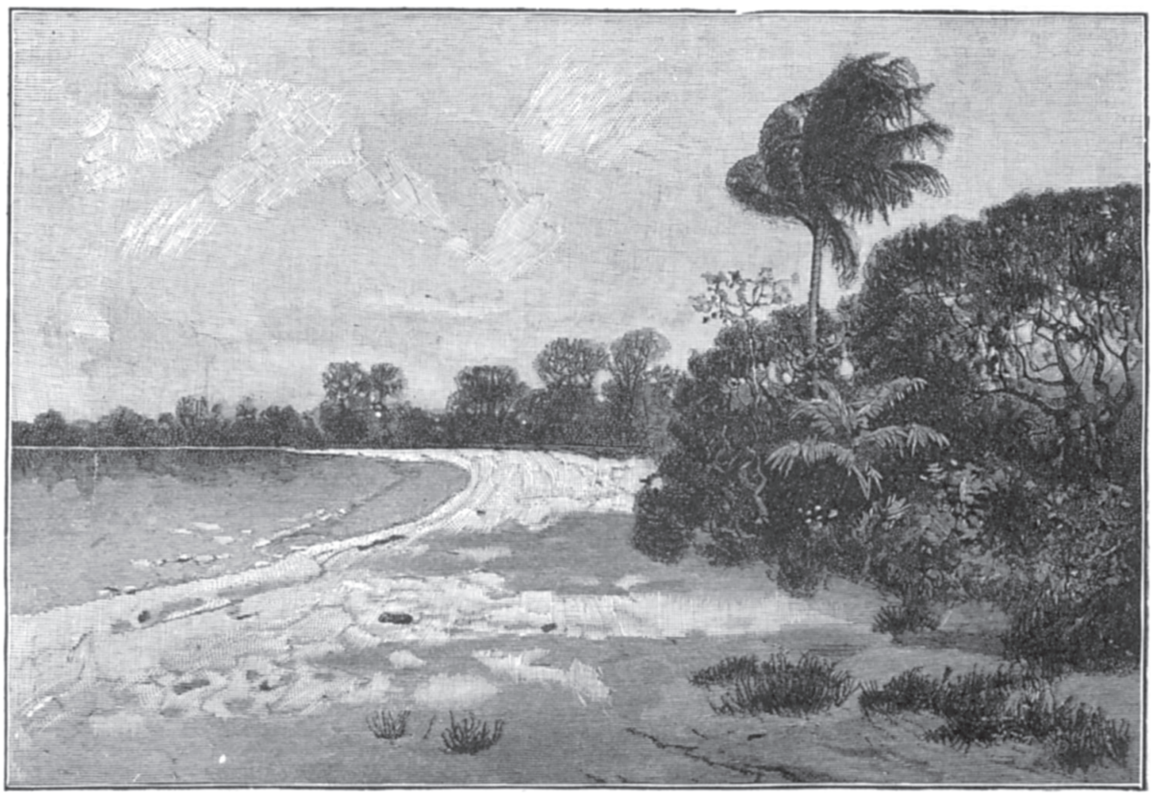

“Caroline island is a genuine atoll, of the type described by Darwin and Dana; the … accompanying illustrations will convey an idea of the scenery of a Pacific atoll. The surface of the island is covered with palms and undergrowth.” From the review of *Memoirs of the National Academy of Sciences* (*The American Naturalist*, 1885, 19:780–781).